

Linear & Logistic Regression

Concepts, Examples, Predictions

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Outline

- 1 Linear Regression
- 2 Logistic Regression
- 3 Comparison & Summary

Linear Regression: Concept

Linear regression models the relationship between a dependent variable Y and one or more independent variables X by fitting a straight line.

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- Y : dependent (predicted) variable X : independent variable
- β_0 : intercept (value of Y when $X = 0$)
- β_1 : slope (change in Y per one-unit change in X)
- ε : random error

Example Dataset

Record	X (Hours Studied)	Y (Exam Score)
1	1	52
2	2	57
3	3	63
4	4	68
5	5	74

We suspect: more hours $\Rightarrow$ higher score.

Compute the Regression Line

Slope and intercept:

$$\beta_1 = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2}, \quad \beta_0 = \bar{Y} - \beta_1 \bar{X}$$

X	Y	X ²	XY
1	52	1	52
2	57	4	114
3	63	9	189
4	68	16	272
5	74	25	370
$\sum X = 15, \sum Y = 314, \sum X^2 = 55, \sum XY = 997$			

$$\beta_1 = \frac{5(997) - 15 \cdot 314}{5(55) - 15^2} = \frac{4985 - 4710}{275 - 225} = \frac{275}{50} = 5.5$$

$$\bar{X} = 3, \bar{Y} = 62.8 \Rightarrow \beta_0 = 62.8 - 5.5(3) = 46.3$$

$$\hat{Y} = 46.3 + 5.5X$$

Predictions (Examples)

Hours X	Predicted Score $\hat{Y} = 46.3 + 5.5X$
0	46.3
1	51.8
2	57.3
3	62.8
4	68.3
5	73.8
6	79.3
7	84.8
8	90.3
9	95.8

Interpretation & Key Insights

- **Slope** = 5.5: each extra study hour increases score by ≈ 5.5 points.
- **Intercept** = 46.3: baseline expected score at $X = 0$.
- Model minimizes the **sum of squared errors**.

Logistic Regression: Motivation

- Linear regression predicts numeric values (e.g., scores).
- For **categorical** outcomes (Pass/Fail, Yes/No, 0/1), we need probabilities in $[0, 1]$.
- **Logistic Regression** maps inputs to probabilities via the sigmoid function.

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

- $P(Y = 1 | X)$: probability of the event (e.g., Pass).
- β_0 : intercept, β_1 : coefficient for X .
- Classify via threshold: if $P \geq 0.5 \Rightarrow$ Pass (1), else Fail (0).

Example Dataset (Binary Outcome)

Record	Hours (X)	Pass (Y)
1	1	0
2	2	0
3	3	0
4	4	1
5	5	1

Students studying fewer than 4 hours tend to fail.

Logistic Function in Action

Assume fitted parameters $\beta_0 = -4.5$, $\beta_1 = 1.2$.

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(-4.5 + 1.2X)}}, \quad z = -4.5 + 1.2X, \quad P = \frac{1}{1 + e^{-z}}$$

Predictions for 10 Examples

Hours (X)	$z = -4.5 + 1.2X$	$P = \frac{1}{1 + e^{-z}}$	Class
0	-4.5	0.011	Fail
1	-3.3	0.035	Fail
2	-2.1	0.109	Fail
3	-0.9	0.289	Fail
4	0.3	0.574	Pass
5	1.5	0.818	Pass
6	2.7	0.937	Pass
7	3.9	0.980	Pass
8	5.1	0.994	Pass
9	6.3	0.998	Pass
10	7.5	0.999	Pass

S-shaped (sigmoid) curve rises steeply after ~ 4 hours.

- Logistic regression outputs **probabilities** (0–1), not a straight line.
- Threshold (commonly 0.5) converts probability to class (Pass/Fail).

Linear vs Logistic Regression

Feature	Linear Regression	Logistic Regression
Output	Continuous (any real)	Probability (0–1)
Equation	$Y = \beta_0 + \beta_1 X$	$P = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$
Error Metric	MSE (Mean Squared Error)	Cross-Entropy / Log Loss
Use Case	Predict numeric values	Classify outcomes (Yes/No)

- Linear regression: simple, interpretable line for continuous outcomes.
- Logistic regression: maps linear combination to a probability via sigmoid; ideal for binary classification.
- Probabilities can be thresholded (e.g., 0.5) for clear-cut predictions.